AL-KAUSAR
COMPUTER INSTITUTE

ADVANCED COURSE

COURSE 8 OF 9

Backend Development

The server side of real systems.

16 weeks	3 × 2 h	96 h	Included
ESTIMATED DURATION	CLASSES PER WEEK	HANDS-ON CLASS TIME	INTERNSHIP + CERTIFICATE

WHAT YOU WILL LEARN

- ✓ Design and build a REST API with validation and error handling.
- ✓ Model data and write queries in PostgreSQL.
- ✓ Implement authentication, roles and common security protections.
- ✓ Write tests and deploy a service with logs.
- ✓ Explain trade-offs in caching, performance and structure.

TOOLS YOU WILL USE

Node.js · Express · PostgreSQL · An ORM (Drizzle or Prisma) · Postman · Git · Docker (intro)

WHO IT IS FOR

People who can already write basic JavaScript and want to build what runs behind websites and apps: orders, users, payments and reports.

BEFORE YOU START

Basic JavaScript. If you have finished our Web Development course, you are ready.

WHERE IT LEADS

Backend developer (junior) · API developer · Database developer · Freelance backend developer

Your course, step by step

10 MODULES · EVERY ONE HAS A HANDS-ON LAB

01 Node.js and server JavaScript — WK 1–2

Running JavaScript outside the browser · Modules, packages and npm · The event loop, async code and promises

02 HTTP and REST APIs — WK 3–4

HTTP methods, status codes and headers · Designing resources and endpoints · Express routing and middleware

03 SQL and PostgreSQL — WK 5–7

Tables, keys and relationships · Normalisation, and when to bend it · SELECT, JOIN, GROUP BY and subqueries

04 ORMs and data access — WK 8

Why use an ORM, and its limits · Schema, relations and queries · Migrations in a team

05 Authentication and security — WK 9–10

Password hashing and storage · Sessions and tokens · Roles and permissions

06 Files, email and payments — WK 11

File uploads and storage · Sending email reliably · Webhooks

07 Testing — WK 12

Unit and integration tests · Testing endpoints and databases · Test data and isolation

08 Caching and performance — WK 13

Where time goes in a request · Caching strategies and Redis basics · Query optimisation

09 Deployment and operations — WK 14

Hosting options and environments · Docker basics · CI: running tests on every push

10 Capstone project — WK 15–16

Choose a real business system · Design the schema and API · Build, test and secure it

YOU FINISH WITH A REAL PROJECT

A production-style API for a real business system, with a PostgreSQL schema, authentication and roles, tests, a CI pipeline, and a deployed, documented service.

INTERNSHIP **Backend intern.** After the course, work with the team on real tasks, with feedback.

FEE AND NEXT BATCH

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CERTIFICATE

Verify online: kataf.pk/verify